

KELLAN MCINTOSH

Data Scientist | Machine Learning | AI Solutions

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Summary

Specialist in Machine Learning, Probabilistic modeling and AI Solutions. Proven experience in architecting RAG-based AI systems on AWS infrastructure to automate complex knowledge retrieval and streamline workflows.

Skills

Python (Pandas, NumPy, Scikit, TensorFlow), SQL, Machine Learning, Probabilistic Modeling (Monte Carlo, Bayesian Inference), Statistical Analysis (Hypothesis testing, regression analysis), AWS, AI Systems (RAG, LLM integration, Bedrock)

Experience

AI for the Commonwealth (AI4CW) program

Amherst, Massachusetts

AI Solutions Architect Intern

01/2026 – Current

- Collaborating with a team to architect and develop a RAG-based AI solution for the Humans Resources department at UMass Amherst.
- Leveraging the use of AWS infrastructure, frontier models and retrieval augmented generation to automate complex HR inquiries.

Massachusetts College of Liberal Arts

North Adams, Massachusetts

Tech Help Desk Consultant

07/2024 - 12/2025

- In my role as an IT Help Desk consultant my responsibilities were to assist students and faculty with various tech related issues they encountered. I assisted with network, MCLA accounts, software and certain hardware issues.
- I have participated in various IT deployments of workstations across campus and assisted with training new staff.

Education

University of Massachusetts Amherst

Amherst, Massachusetts

B.Sc. Computer Sciences

01/2026 – Current

Massachusetts College of Liberal Arts

North Adams, Massachusetts

B.Sc. Computer Sciences

01/2024 – 12/2025

Projects

Berkshire Bounty Data Report

Collaborated with a team to analyze the biannual Berkshire Bounty survey. I analyzed diverse funding sources for regional food pantries to identify counties facing critical funding gaps. I investigated patterns correlating funding shortages with specific missing sources of income.

MRI Tumor Detection Model

- Engineered a Convolutional Neural Network (CNN) to automate the detection of various brain tumor types from high-resolution MRI datasets.
- Implemented dropout regularization techniques to mitigate overfitting, achieving a 93% classification accuracy on unseen test data.